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Performance Analytics Dashboard

Kimia Farma Business Performance Analysis 2020-2023

Kimia Farma - Big Data Analytics

Presented by
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Rehana Putri Salsabilla

An Information Technology student at the Sepuluh Nopember Institute of Technology with a career interest in Data Analyst. Experienced in data processing, analysis, and dashboard creation to generate business insights. Experienced in using tools such as SQL (BigQuery), Looker Studio, and basic data processing to support decision-making.

Education :

- Bachelor of Information Technology student, ITS
- Data Analyst Bootcamp - Celerates CAMP Batch 4

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Courses and Certification

Bootcamp Data Analyst with SQL & Python , DQLab | Sertifikat	**<August, 2025>**
Bootcamp Machine Learning & AI , DQLab | Sertifikat	**<August, 2025>**
Bootcamp Data Analyst with Excel , DQLab | Sertifikat	**<August, 2025>**
Data Science Introduction , MySkill | Sertifikat	**<July, 2025>**
Intro to Data Analyst , Coding Studio | Sertifikat	**<July, 2025>**
Intro to Data Analytics , RevoU | Sertifikat	**<July, 2025>**

About Company

Kimia Farma was the first pharmaceutical company in Indonesia, founded in 1817 by the Dutch East Indies government under the name NV Chemicalien Handle Rathkamp & Co. After independence, the company was nationalized and officially became PT Kimia Farma (Persero) in 1971. In 2001, Kimia Farma became a public company and was listed on the Indonesia Stock Exchange.

Along with its development, Kimia Farma has become part of the State-Owned Pharmaceutical Holding under PT Bio Farma (Persero) since 2020. To date, Kimia Farma has developed as an integrated healthcare service company that includes production, distribution, and a network of pharmacies and clinics in various regions of Indonesia.

Business Problem & Objective

Business Problem

Kimia Farma's sales performance for the 2020–2023 period exhibited a relatively fluctuating pattern, although overall it tended to be stable. This indicates differences in performance across time, regions, and branches. Furthermore, there is a potential gap between branch service quality and customer transaction experience, as well as the impact of discount strategies on profitability, which require further analysis.

Objective

- Analyze sales and profit trends for the 2020–2023 period
- Identify top-performing branches and provinces
- Identify gaps between branch ratings and transaction ratings
- Evaluate the impact of discounts on profits
- Provide insights to support business decision-making

Project Portfolio

Overview

This project is part of the Big Data Analytics challenge, which aims to analyze Kimia Farma's business performance for the 2020–2023 period using available transaction data.

The challenge involves an end-to-end analysis process, from data processing in BigQuery to creating analysis tables and visualizing them in the form of an interactive dashboard in Looker Studio, generating insights that can support business decision-making.

Challenge

- Process data using BigQuery
- Create analytical tables from multiple datasets
- Build dashboards in Looker Studio
- Present business insights from analysis results

Tools used: BigQuery • Looker Studio • SQL

Work Pipeline

Data Ingestion

Importing a dataset into BigQuery as a data source

Data Understanding

Understanding data structures, columns, table relationships

Data Processing

Performing joins and calculations using SQL BigQuery

Data Transformation

Create a kf_analysis analysis table for dashboard needs

Data Visualization

Building an interactive dashboard using Looker Studio

Reporting

Presenting insights in the form of dashboards and presentations

Entity Relationship Diagram

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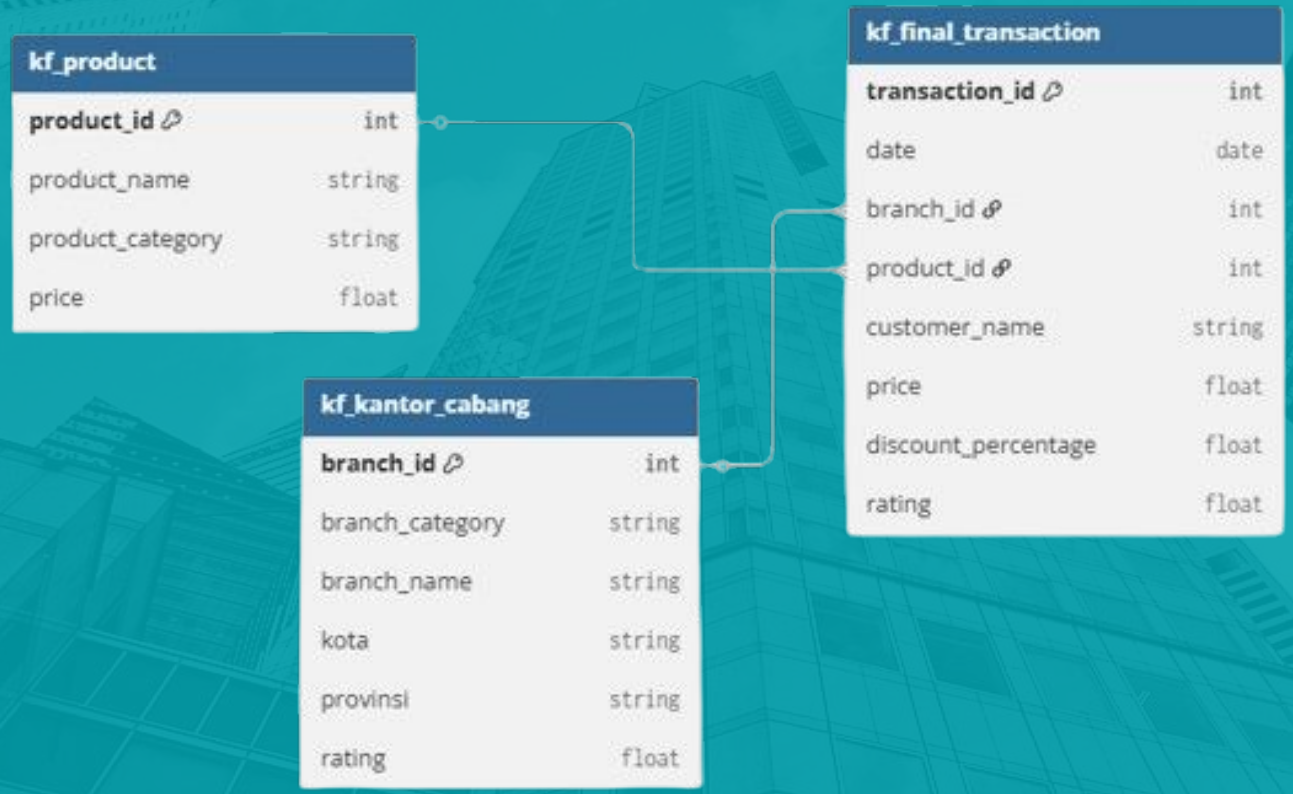

Data Understanding

Dataset	Description	Main Information
kf_final_transaction	Sales transaction data	Date, sales, profit, discount, transaction rating
kf_product	Product information	Product name, category, price
kf_kantor_cabang	Branch & location information	Branch name, city, province, branch rating
Kf_analysis (Master)	Processed data	Combined data for dashboard analysis purposes

Total data includes ±670,000 transactions for the 2020–2023 period which are used for business performance analysis.

Data processing & Transformation

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The data is processed using SQL in BigQuery by combining transaction, product, and branch datasets to produce an analysis table (kf_analysis) used in the dashboard.

```sql
CREATE OR REPLACE TABLE
`manifest-alpha-491205-e6.kimia_farma.kf_analysis`
AS

SELECT
    t.transaction_id,
    t.date,
    t.branch_id,
    c.branch_name,
    c.kota,
    c.provinsi,
    c.rating AS rating_cabang,
    t.customer_name,
    t.product_id,
    p.product_name,
    t.price AS actual_price,
    t.discount_percentage,

    -- PERHITUNGAN GROSS PROFIT (%)
    CASE
        WHEN t.price <= 50000 THEN 0.10
        WHEN t.price <= 100000 THEN 0.15
        WHEN t.price <= 300000 THEN 0.20
        WHEN t.price <= 500000 THEN 0.25
        ELSE 0.30
    END AS persentase_gross_laba,

    -- PERHITUNGAN NETT SALES
    t.price - (t.price * t.discount_percentage) AS nett_sales,

    -- PERHITUNGAN NETT PROFIT
    (t.price - (t.price * t.discount_percentage)) *
    CASE
        WHEN t.price <= 50000 THEN 0.10
        WHEN t.price <= 100000 THEN 0.15
        WHEN t.price <= 300000 THEN 0.20
        WHEN t.price <= 500000 THEN 0.25
        ELSE 0.30
    END AS nett_profit,

    -- RATING TRANSAKSI
    t.rating AS rating_transaksi

FROM `manifest-alpha-491205-e6.kimia_farma.kf_final_transaction` AS t

LEFT JOIN `manifest-alpha-491205-e6.kimia_farma.kf_kantor_cabang` AS c
    ON t.branch_id = c.branch_id

LEFT JOIN `manifest-alpha-491205-e6.kimia_farma.kf_product` AS p
    ON t.product_id = p.product_id;
```

Row	transaction_id	date	branch_id	branch_name	kota	provinsi	rating_cabang	customer_name	product_id
1	TRX5758357	2023-07-20	83742	Kimia Farma - Apotek	Pematangsiantar	Sumatera Utara	4.8	Robin Brown	KF172

product_name	actual_price	discount_percentage	persentase_gross_laba	nett_sales	nett_profit	rating_transaksi
Psycholeptics drugs, Hypnotics and sedatives drugs	2100	0.15	0.1	1785.0	178.5	3.1
Psycholeptics drugs, Hypnotics and sedatives drugs	2100	0.15	0.1	1785.0	178.5	3.5

Dashboard Design

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Performance Analytics Kimia Farma 2020–2023

Profit Margin: 28,4%

Total Net Sales: 321,2 M

Total Net Profit: 91,2 M

Avg Transactions Rating: 4,0

Total Transactions: 672,5 rb

Branch Name | Provinsi | Year

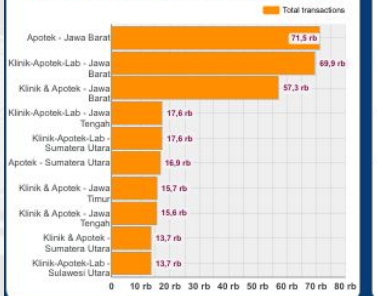
Performance Summary Table

Year	Total Transactions	Total Net Sales	Total Net Profit
2020	168,7 rb	80,4 M	22,8 M
2021	167,7 rb	80 M	22,7 M
2022	168,6 rb	80,6 M	22,9 M
2023	167,5 rb	80,1 M	22,8 M

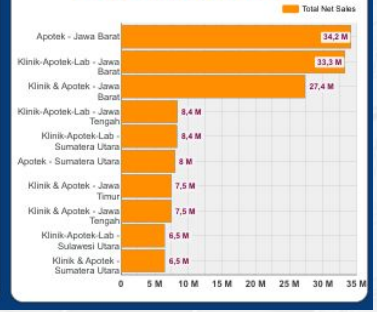
Net Sales Trend 2020 – 2023

* Sales showed a stagnant trend during 2020–2023 without significant growth.

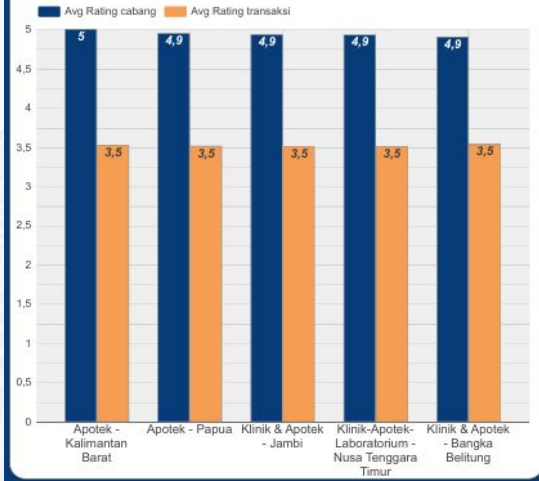
Top 10 Branch by Total Transactions



Top 10 Branch by Nett Sales



Top 5 High Branch Rating but Low Transactions Rating



Profit Distribution by Province

* Most of the profits come from West Java, indicating a high dependence on this region.

	Province	Total Net Profit
1.	Jawa Barat	26,9 M
2.	Sumatera Utara	6,5 M
3.	Jawa Tengah	6,3 M
4.	Jawa Timur	4,7 M
5.	Sulawesi Utara	4,5 M
6.	Sumatera Barat	4,3 M
7.	Aceh	3 M
8.	Nusa Tenggara Barat	2,8 M
9.	Kalimantan Timur	2,8 M
10.	Riau	2,7 M

Impact of Discount on Average Profit

* The higher the discount given, the lower the profit obtained.

Insight Business Performance Trend (2020–2023)

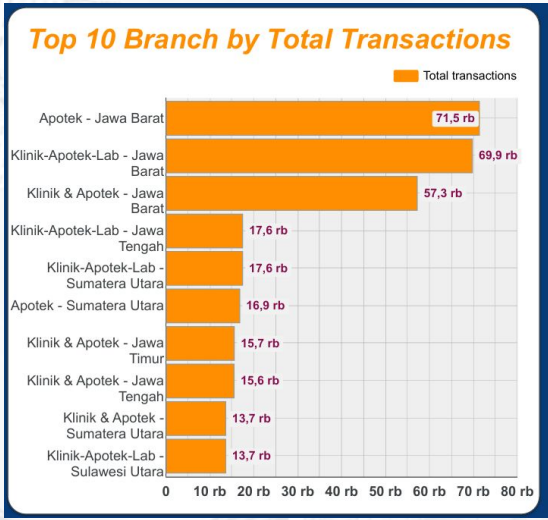
Key Performance Insight

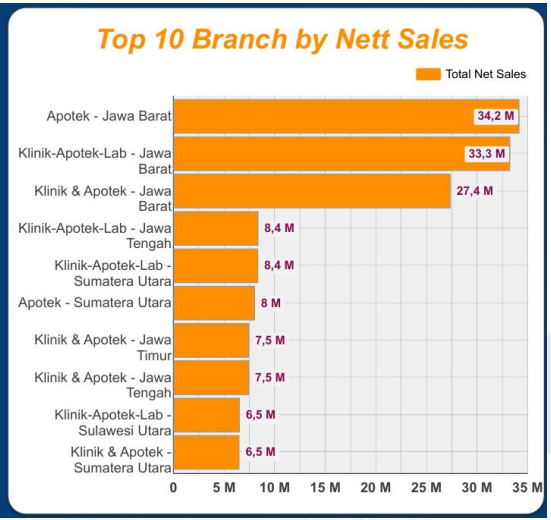
Kimia Farma's sales have been relatively stable at around 80 million annually, with a decline in 2021, an increase in 2022, and a decline again in 2023. Overall, no significant growth has been observed, thus concluding that business performance is likely stagnant.

Business Analysis

This situation indicates that despite stable business operations, the company has not been able to drive sustainable growth. This indicates the need for a more effective strategy to increase sales and encourage future business expansion.

Insight Branch Performance Analysis





Key Performance Insight

Based on the number of transactions, the West Java branch had the highest transaction volume compared to other branches, indicating very high sales activity. This aligns with the results for total net sales, where the same branch also generated the highest sales value, thus concluding that the branch is not only busy but also contributes significantly to revenue.

Business Analysis

The high transaction volume and sales value indicate that the branch's performance is quite optimal in converting sales activity into revenue. However, the significant difference compared to other branches indicates that business performance remains uneven, necessitating strategies to increase the contribution of branches in other regions.

Insight Geographical Profit Distribution

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Profit Distribution by Province

* Most of the profits come from West Java, indicating a high dependence on this region.

	Province	Total Net Profit
1.	Jawa Barat	26,9 M
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7.	Aceh	3 M
8.	Nusa Tenggara Barat	2,8 M
9.	Kalimantan Timur	2,8 M
10.	Riau	2,7 M

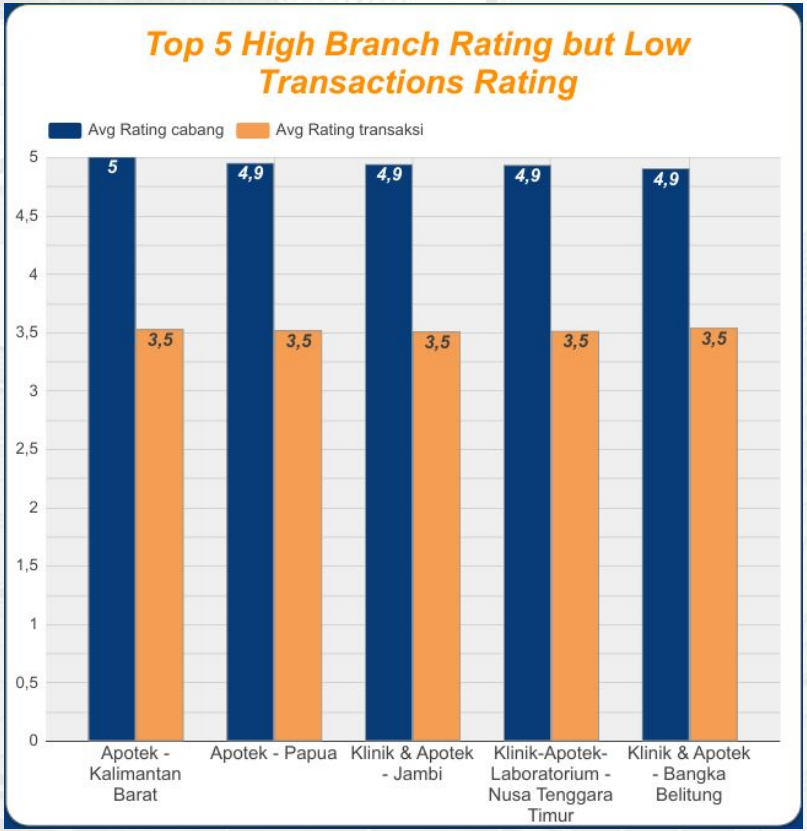
Key Performance Insight

Profit distribution shows that the majority of profits come from West Java province, with a significantly higher value than other regions. This significant difference indicates that profit contributions are uneven and remain concentrated in one key region.

Business Analysis

This situation indicates a high level of business dependence on West Java, which could pose a risk if performance in that region declines. It also indicates that market potential in other regions is underutilized, necessitating expansion strategies and performance improvements outside the core region.

Insight Customer Experience Analysis



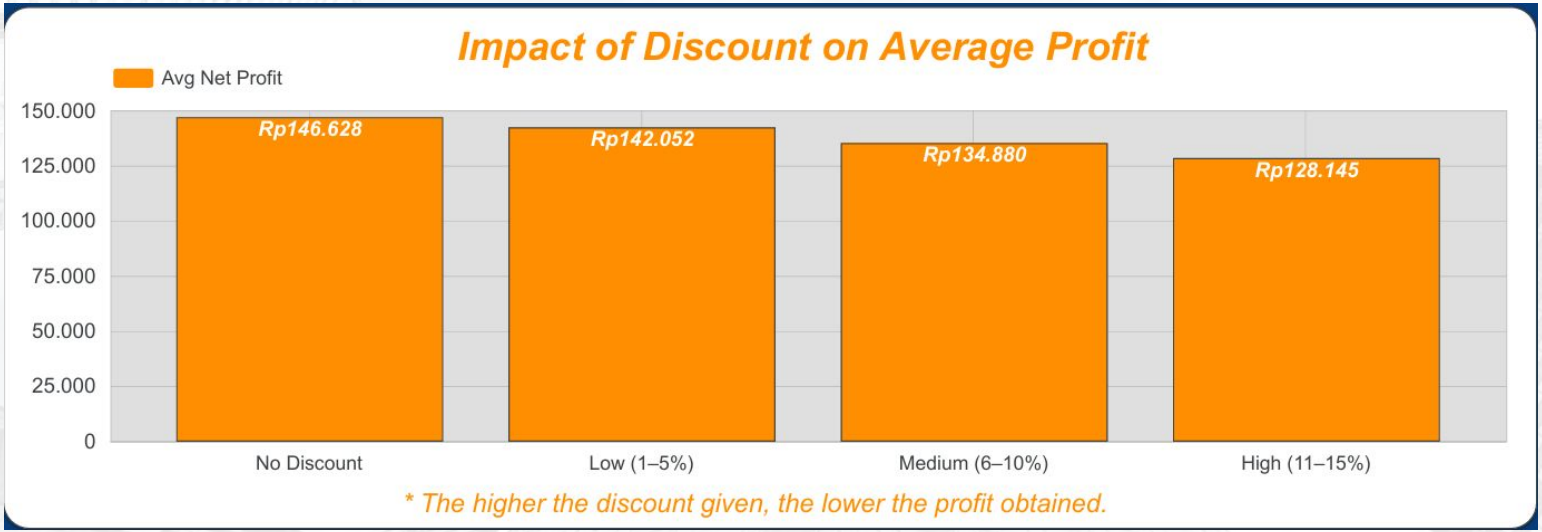
Key Performance Insight

Based on the rating analysis, there is a significant difference between branch ratings and transaction ratings, with branches receiving very high ratings (around 4.9–5), but customer transaction ratings tending to be lower (around 3.5). This indicates a mismatch between perceived branch quality and the actual customer experience during transactions.

Business Analysis

This situation indicates potential operational issues, such as transaction processing, employee service, or suboptimal service systems. Although the branch's facilities and image are good, the customer experience during in-person interactions may not fully meet expectations, necessitating evaluation and improvement of service quality.

Insight Impact of Discount on Profitability



Key Performance Insight

Based on the analysis, the highest average profit was achieved on products without discounts, and it consistently decreased as the discount level increased. This indicates a negative relationship between discounts and profits, where the larger the discount, the lower the profit.

Business Analysis

This situation indicates that the current discount strategy is suboptimal and has the potential to reduce business profitability. Excessively high discounts don't always provide comparable benefits, so evaluation is necessary to ensure more targeted discount use and maintain a balance between increasing sales and profits.

BUSINESS RECOMMENDATION

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Drive Sales Growth Strategy

Focus on increasing sales growth by developing more aggressive marketing strategies, such as customer-driven promotions and product expansion in branches with stagnant performance. This is crucial to address stable business conditions but the lack of significant growth during the 2020–2023 period.

Improve Service & Operational Quality

Improve service and operational quality at branches with gaps between branch ratings and transaction ratings, particularly in customer service and transaction processing. These improvements can increase customer satisfaction while supporting overall business performance.

Reduce Regional Dependency

Reduce dependence on West Java by optimizing the performance of branches in other provinces through expansion strategies, increased marketing activities, and product adjustments to local market characteristics. This aims to create a more equitable distribution of profits and reduce business risk.

Optimize Discount Strategy

Evaluate and limit discounts, especially in low-profit categories, and focus on a more targeted, data-driven discount strategy. This is crucial to maintaining a balance between increasing sales and profitability, as high discounts have been shown to reduce profits.

CONCLUSION

Conclusion	Details
Stable but No Significant Growth	Kimia Farma's business performance during 2020–2023 was relatively stable, but did not show significant growth, so a strategy was needed to drive increased sales.
High Dependency on Certain Regions	Business performance is still concentrated in certain regions, especially West Java, which indicates an imbalance in contribution and potential risk of dependency.
Operational and Strategy Gaps	There are gaps in service quality and discount strategies that have an impact on decreasing profits, so operational improvements and strategy evaluations are needed.
Need for Strategic Improvement	A more optimal strategy is needed to increase growth, equalize performance, and business efficiency to achieve better results.

Thank You

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